Project - Deliverable 1

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Sheridan College

SYST17796 - Fundamentals of Software Design and Development

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Table of Contents

1. Project Background and Description —---------- 3
2. Project Scope —---------- 5
3. High Level Requirements —---------- 6
4. Implementation Plan —---------- 7
5. Design Considerations —---------- 8
6. Starter Code UML —---------- 9
7. Group Contract —---------- 10

Project Background and Description

The goal of this project is to create a fully functional and multiplayer version of the card game “Go Fish” using Java.

Rules of Game:

1. The game must consist of 2-5 players, and uses a single deck of cards with Jokers removed (52 cards).
2. Each player starts with 5 cards in their hand. The remaining cards are in the draw deck, face-down.
3. Each player rolls a dice, the player with the highest number goes first.
    - Player turns are sequenced clock-wise.
4. On player’s active turn:
    - They can ask ANY other player for a specific value of card (e.g. 8).
        - If the other player has this value in their hand, they must hand over all cards of that value to the active player.
        - If the other player does not have this value, they say, “Go fish”. The active player will then draw 1 card from the deck.
    - The active player can create “books” containing 4 cards of equal value from their hand.
        - Books must be removed from the player's hand and placed on the table face-up.
5. The game is over when any player has no more cards remaining in their hand.
6. The winner is the player who ended the game with the highest amount of “books”.

Starter Code Analysis:

The starting code includes four .java files:

- Card.java
- Game.java
- GroupOfCards.java
- Player.java

So far, the project is made up of abstract classes with basic card game structure but without specific behaviours and methods designed for a particular game. The data members are private and are accessed through accessors and mutators. In GroupOfCards.java, the cards list is not initialized in the constructor. This would cause a NullPointerException when calling certain methods such as shuffle() or getCards(). Overall, the starting code is quite incomplete in both game logic.

Project Scope

Group Members and Roles:

Ke Xin (Christina) Gao (student number: 991610458)

I am the sole member of this group, and I will complete the entire scope of the project.

Technical Scope:

This game will run from the output of a console. Player turns will be executed through text input, and the game state will also be displayed through text formatting. Since there will only be a singular console shared amongst all the players, an interface will be created to prompt for player turns and show only the necessary information to the active player. Additionally, during all player turns, they will be able to see the draw deck size, number of cards in other players' hands, the number of "books" attributed to each player, and their own hand.

The project will be considered completed after the following criteria has been met:

- A custom game size of 2, 3, 4, or 5 players can be created.
- The rules are understandable to the players involved.
- The rules outlined in Project Background are implemented and verified through tests (e.g. invalid inputs are not accepted).
- Classes within this project follow OOP principles and demonstrate loose coupling, high cohesion.
- UML diagrams depict the Java code correctly.
- All files are committed to the repository.
- The full game can be played from start to finish without errors and crashes.

High-Level Requirements

Go Fish new system requirements:

1. Ability for 2-5 players to register in a “game lobby” with unique player names.
2. Ability to create and shuffle a deck of 52 cards and deal an appropriate amount of cards to each player.
3. Ability for each player to roll a die, and reroll upon ties.
4. Ability for the active player to target any other player during their turn.
5. Ability for the active player to ask for a card of any value.
6. Ability for the system to check for card value in targeted player’s hand and transfer cards or signal “Go fish”.
7. Ability to detect 4 cards of the same value in a player’s hand and remove them upon player request.
8. Ability to record player-created “books” on the playing board.
9. Ability for players to draw from the draw deck.
10. Ability to advance turns clockwise after the active player declares “End turn”.
11. Ability for the active player to see all appropriate information regarding the game state and their own hand.
12. Ability to detect the end of the game, calculate the winner, and declare the winner.

Implementation Plan

Project Github:

https://github.com/christinakgao/syst17796-project

Comments are expected with each commit. Pushes to the remote repository are to be made at the end of a working day after reviews are made on any changes.

Notable directories in the project include:

- /src - for all Java course code
- /docs - PDFs, text, and word documents related to the project (such as Group Contract and Design Template)
- /uml - UML diagrams

Tools expected for this project:

- NetBeans
- Visual Paradigm

Coding standards to follow:

- OOP principles
- Loose coupling
- High cohesion
- Header comments that include:
    - Contributing member names
    - Title of project
    - Date of project
    - Summary of project
- Project comments that are concise and accurately provide information on methods/variables

Design Considerations

Encapsulation:

- The current starter code has private data members.
    - E.g. Player.java has data member ‘private String name;’ and method ‘public String getName()’.
- The starter code is currently structured with abstract classes.
- For this project, I will likely create additional classes such as ‘Hand.java’ in order to provide a controlled method to modify cards in players’ hands.
- Need to create validation for player names and player inputs (not null, fits the parameters of the game).

Delegation:

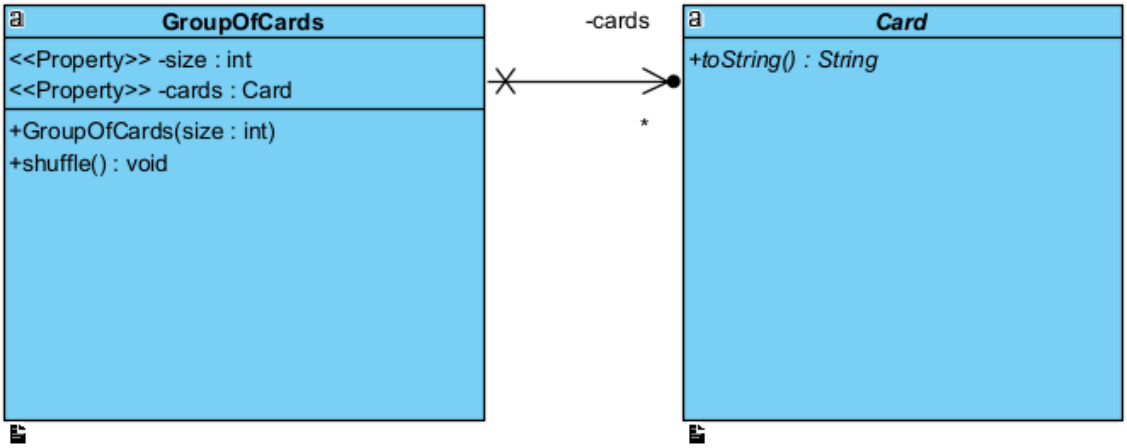
- Game.java delegates to Player.java. The Game.java class stores the players and allows each player to individually implement actions within the game.
- Currently the shuffling is delegated to GroupOfCards.
- Formatting will be delegated to each class’ toString(), however, console output will be a separate class on its own to ensure that game logic is not mixed up with game display.

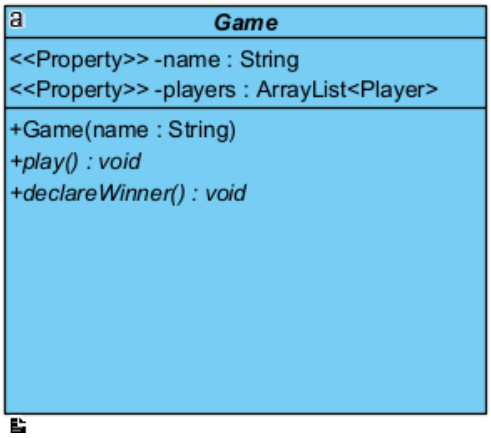
Flexibility/Maintainability:

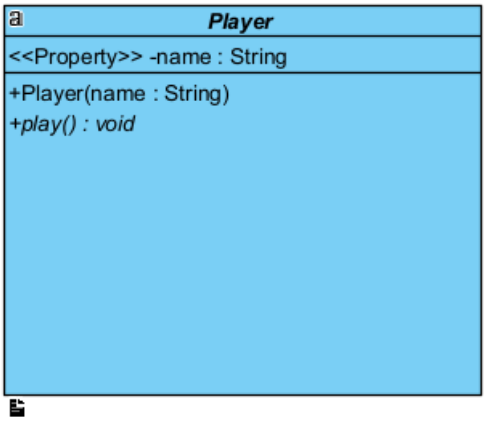
- Since the base classes Card.java, Game.java, and Player.java are abstract, they are reusable in other game formats. This game will extend these abstract classes rather than modify them.
- GroupOfCards.java also has flexibility in its size, which allows for it to be used in a multitude of card games that have decks of different sizes.
- Need to have data members like Suit and Value represented with enums and constants rather than lists of hard-coded Strings.

Starter Code UML Diagram

UML Diagram files are also uploaded onto the Git repository.







Team Contract

SYST 17796 TEAM PROJECT

Team Name: CKG

Please negotiate, sign, scan and include as the first page in your Deliverable 1.

Please note that if cheating is discovered in a group assignment each member will be charged with a cheating offense regardless of their involvement in the offense. Each member will receive the appropriate sanction based on their individual academic integrity history.

Please ensure that you understand the importance of academic honesty. Each member of the group is responsible to ensure the academic integrity of all of the submitted work, not just their own part. Placing your name on a submission indicates that you take responsibility for its content.For further information, read Academic Integrity Policy here : https://caps.sheridancollege.ca/student-guide/academic-policies-and-procedures.aspx

Team Member Names (Please Print)	Signatures/Initials	Student ID
Project Leader: Ke Xin (Christina) Gao		991610458

By signing this contract, we acknowledge having read the Sheridan Academic Integrity Policy

Responsibilities of the Project Leader include:

- Assigning tasks to other team members, including self, in a fair and equitable manner.
- Ensuring work is completed with accuracy, completeness and timeliness.
- Planning for task completion to ensure timelines are met.
- Notifying the professor of any issues in a timely manner so that corrective measures can be taken.
- Any other duties as deemed necessary for project completion.

What we will do if . . .

Scenario	**Accepted initials**	**We agree to do the following (Put an X corresponding to your choice in each box)**
Team member does not regularly attend team meetings and/or does not respond to communications in a timely manner.		Project leader emails the student citing the concerns and cc's the professor so they are aware of the situation at the very onset <u>X</u> (**Mandatory**). a) __ In addition to above, the leader/team will (add your own content here):
Team member does not deliver component on time due to severe illness or extreme personal problem.		a) Team absorbs workload temporarily <u>X</u> b) Team seeks advice from professor __ c) Team shifts target date if possible __ d) __ Other (specify):
Team member has difficulty delivering component on time due to lack of understanding or ability.		a) Team reassigns component __ b) Team helps member __ c) Team member must ask professor for help <u>X</u> d) __ Other (specify):

Scenario	**Accepted initials**	**We agree to do the following (Put an X corresponding to your choice in each box)**
Team member does not deliver component on time due to lack of effort.		a) Team absorbs workload __ b) Team member(s) ask professor to request a Participation Form from <u>all</u> team members. This *may* result in individualized grades being awarded for a deliverable __ c) Both a. and b. above <u>X</u> d) __ Other (specify):
Team cannot achieve consensus leaving one or more member(s) feeling that their voice(s) is/are not being heard in a decision which affects everyone.		a) Team agrees to abide by majority vote <u>X</u> b) Team seeks advice from the professor __ c) __ Other (specify):
Team members do not share expectations for the quality of work on a particular deliverable.		a) Team members will draw on each other's strengths to help bring the quality of the deliverable to a minimal acceptable level <u>X</u> b) Team votes on each submission's quality __ c) Team member(s) ask professor to request a Participation Form from all team members, which may result in individualized grades being awarded for a deliverable __ d) __ Other (specify):
Team member behaves in an unprofessional manner, e.g. being rude, uncooperative and/or making one or more		a) Team agrees to avoid use of all vocabulary inappropriate to a business/college setting __

Scenario	Accepted initials	We agree to do the following (Put an X corresponding to your choice in each box)
member(s) feel uncomfortable.		b) Team attempts to resolve the issue by airing the problem at a team meeting __ c) Team requests a meeting with the professor to discuss further X d) __ Other (specify):
There is a dominant team member who insists on making all decisions on the team's behalf leaving some team members feeling like subordinates rather than equal members		a) Team will actively solicit consensus on all decisions which affect project direction by asking for each member's decision and vote X b) Team will express subordination feelings and attempt to resolve issue __ c) Team seeks advice from the professor __ d) __ Other (specify):
Team has a member who refuses to participate in decision making but complains to others that s/he wasn't consulted		a) Team forces decision sharing by routinely voting on all issues __ b) Team routinely checks with each other about perceived roles __ c) Team discusses the matter at team meeting X